

ANALOG & DIGITAL CIRCUITS

Lecture 8

Computer Science Department,
TUCN

Positive feedback circuits: MULTIVIBRATOR CIRCUITS

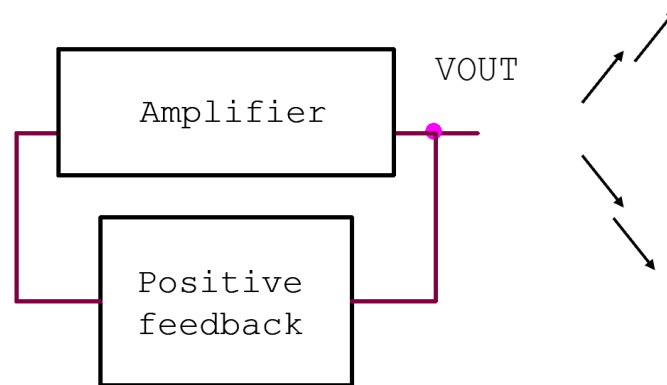
Bistable

Monostables

Astables

Multivibrators

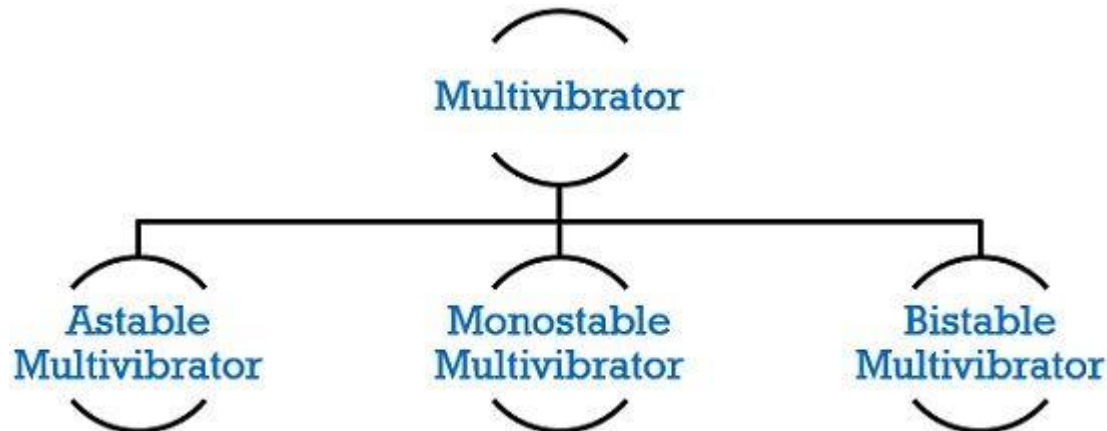
- Circuits characterized by the existence of some **well-defined states**, amongst which take place **fast transitions**, called **switching processes**.
- A **switching process** is a fast change in value of a current or a voltage, the fast process implying the existence of **positive reaction loops**, or negative resistances.
- The switching can be **triggered** from **outside**, by means of command signals, or from **inside**, by slow charge accumulation and the reaching of a critical state by certain electrical quantities in the circuit.



Remember Positive
Reaction (feedback) circuits

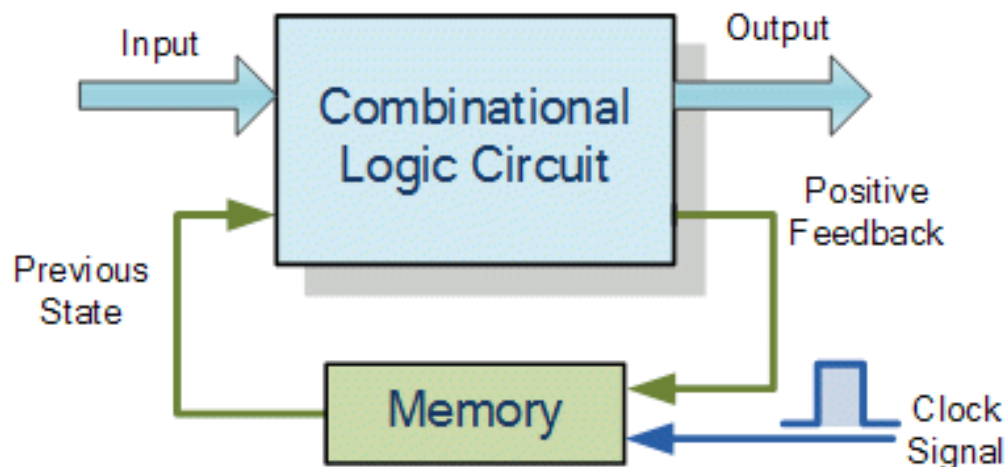
$$V_{out} \text{ (at time } t_{N+1}) = V_{in} + V_{out} \text{ (at time } t_N)$$

- Circuits have two, well defined states, which can be either stable or unstable
 - A **stable state** is a state, in which the circuit, in absence of a driving signal, can remain for an unlimited period of time
 - The circuit can remain in an **unstable state** only for a limited period of time, after which, in the absence of any exterior command signals, it switches into the other state.
- The multivibrator circuits can be grouped, according to their number of stable (steady) states, into:
 - **bistable(flip-flops) circuits** with both states being stable
 - **monostable circuits** having a stable and an unstable state
 - **astable circuits** with both states being unstable



Bistable (flip-flop) circuits

- The main feature of the flip-flop circuits is the existence of two stable states, in which the circuit may remain for a long time. The switching from one state to the other is triggered by command signals
- Flip flop: an example for a **sequential circuit** (a circuit with outputs that present logical values depending on a certain sequence of signals, that have previously existed in the circuit). Because of this behavior, sequential circuits have the capability of **storing information (memories)**.
 - Unlike sequential circuits, **combinational circuits**, consisting of logical gates, have outputs, which depend only on the current inputs
- Flip-flop circuits may be classified into **symmetrical** and **non-symmetrical**

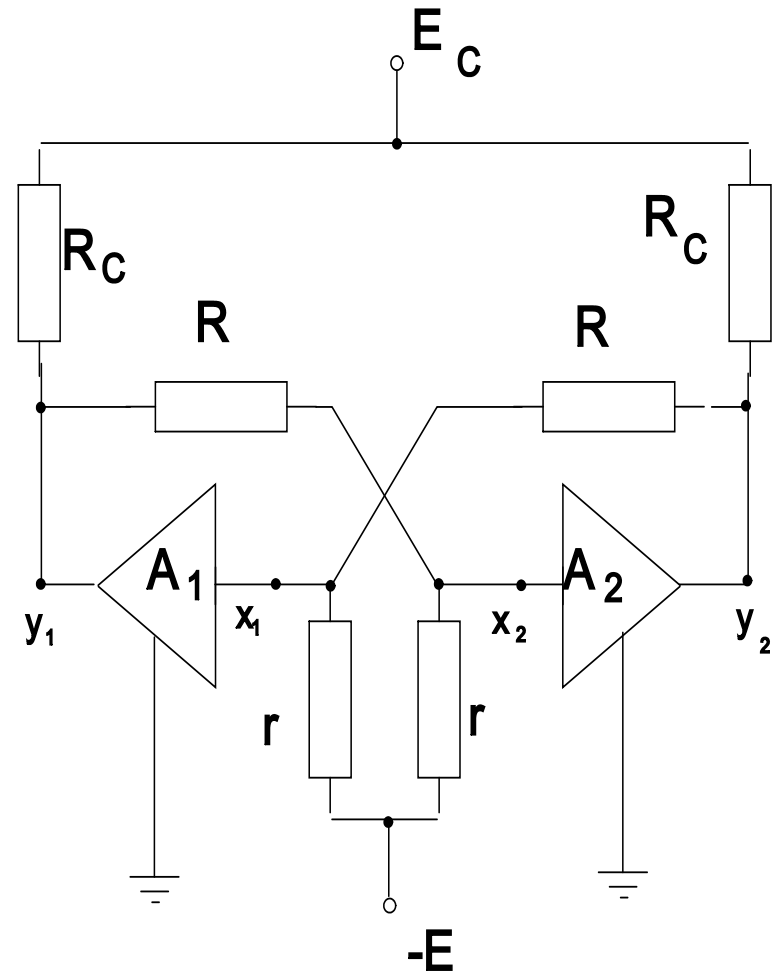


Sequential
Logic Circuits

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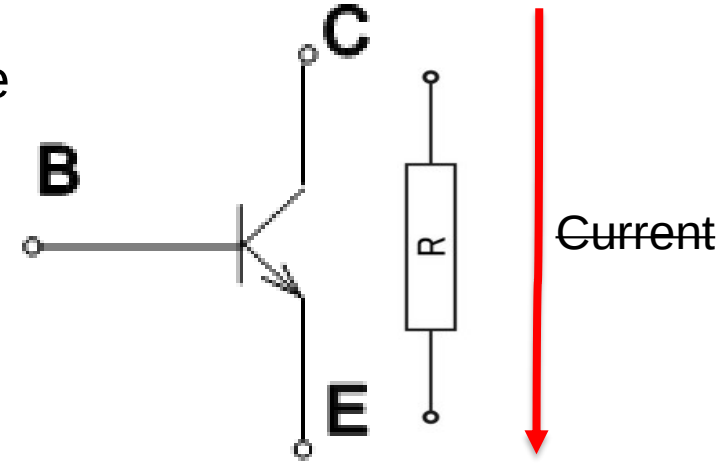
Symmetrical Flip-flop Circuits With Discrete Components

- Basic layout presented at right
- A_1 and A_2 are two amplifiers connected in a positive reaction loop, through the voltage divider made out of the resistances R and r



A. If the BJT is closed ($V_{BE} < 0,6V$)

$\Rightarrow R_{CE}$ has a very high value and V_{CE} can have different values, depending on other components in the circuit (can have large values)



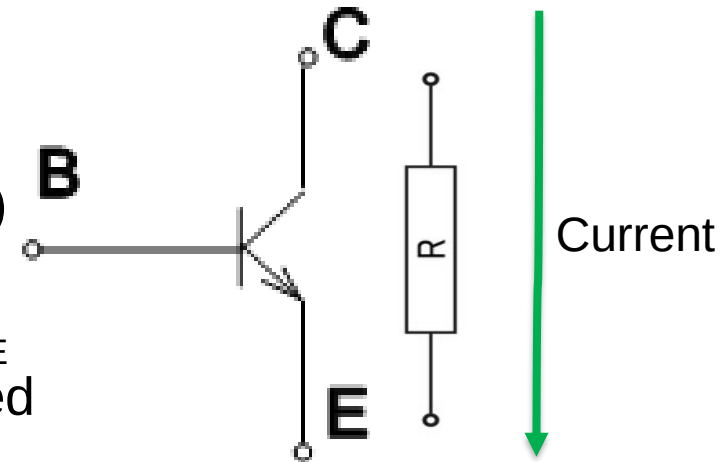
B. If the BJT is opened

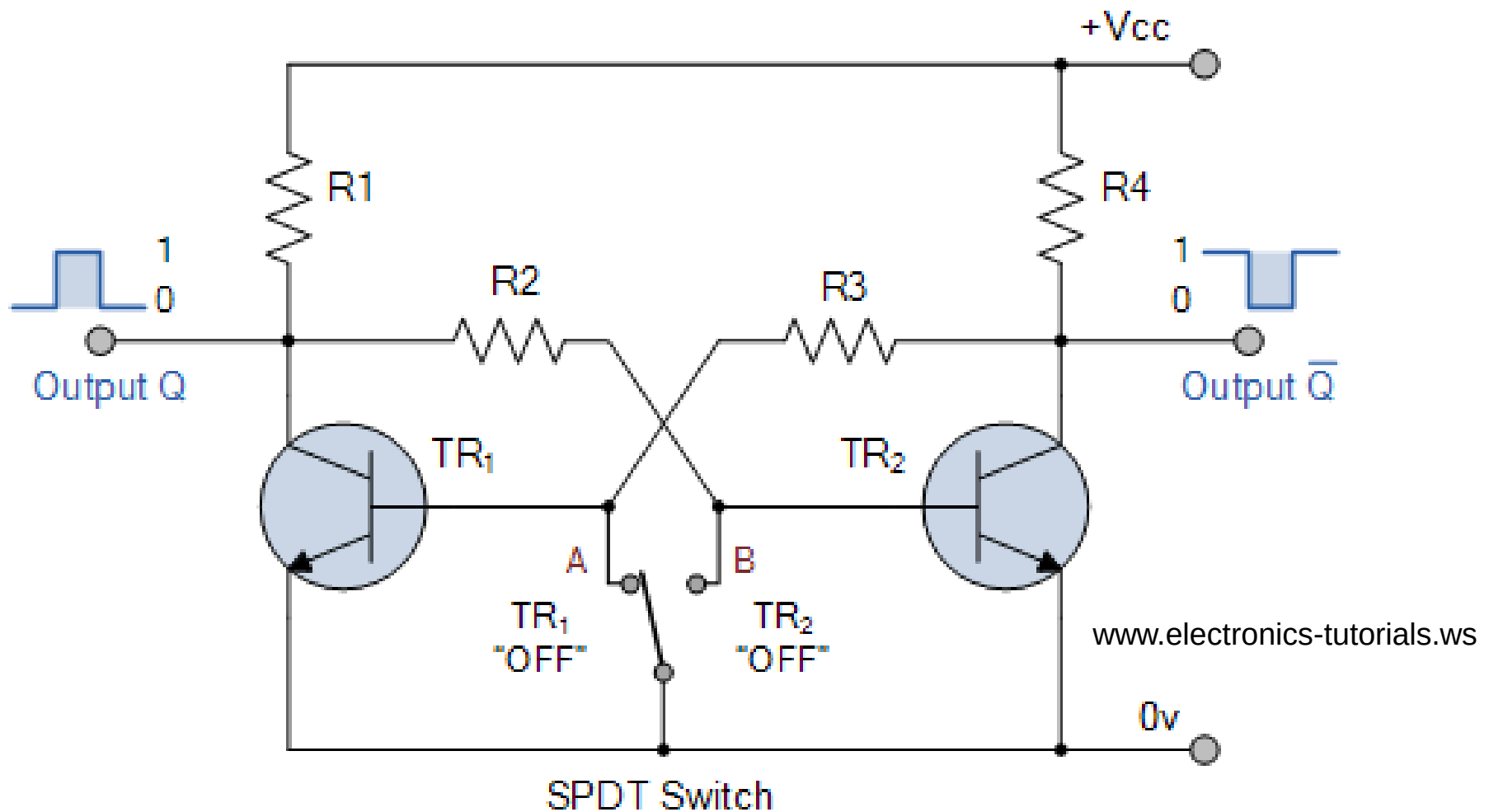
B1. $V_{BE} = 0,6V$

$\Rightarrow R_{CE}$ value will decrease (however, still high value) and the current will flow between C and E; V_{CE} can have large values

B2. V_{BE} between 0,6V and 0,75V (V_{BE} will be limited to 0,75V so not to damage the junction)

$\Rightarrow R_{CE}$ value will be small $\Rightarrow$ the current between C and E will increase its value; V_{CE} will drop; when the BJT is completely opened (at saturation) V_{CE} will drop to its minimum value: $V_{CEs} \approx 0,2V$



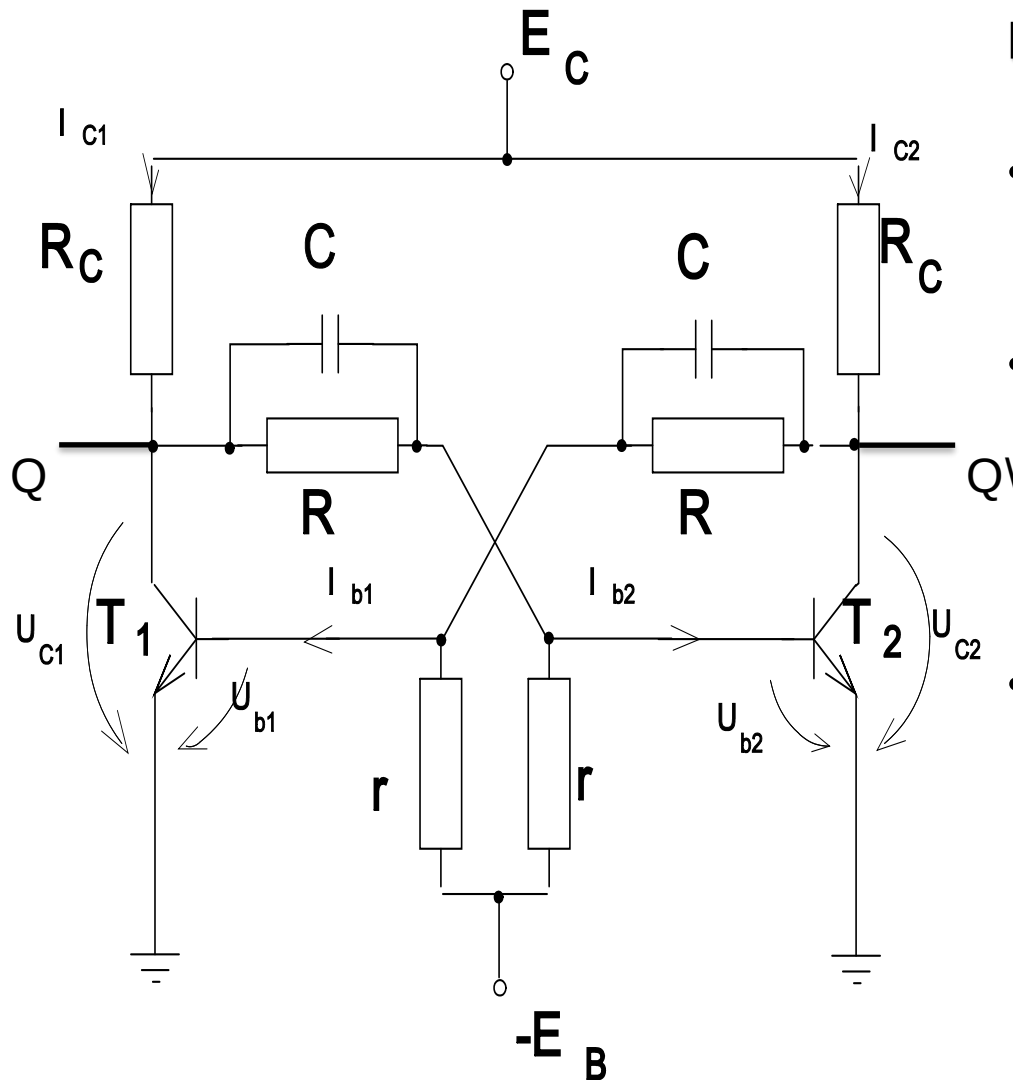


If switch is on A position =>

- TR_1 is cut-off (base driven to ground) and $Q = 1$ (from V_{CC}) just like in the case of BJT inverter
- TR_2 is saturated (base driven by V_{CC} through R_1 and R_2) and $\bar{Q} = 0$ ($V_{CE} \sim 0,2V$) just like in the case of BJT inverter

If switch is on B position =>

- TR_1 and TR_2 will switch so $Q=0$ and $\bar{Q}=1$



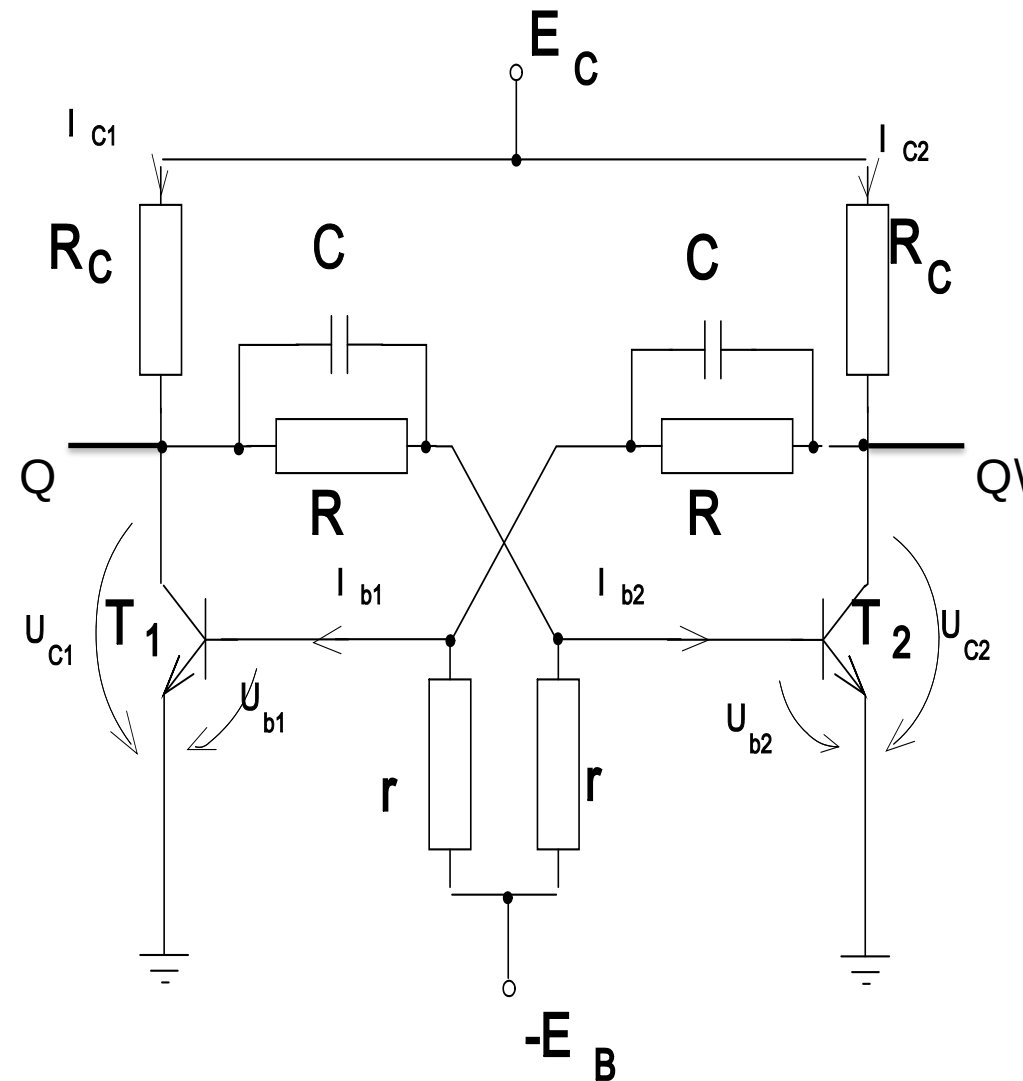
Design with discrete components

- Amplifying stages made out of the transistors T_1 and T_2
- Connected together through the positive reaction represented by the voltage dividers R and r .
- A part of the collector-emitter voltage of one transistor is transmitted into the base of the other transistor

Assume an initial state:

$$Q=1 \text{ and } Q\|=0 \Rightarrow$$

T_1 cut-off and T_2 saturated



Assume a command signal so that I_{C1} has a small increase

$$I_{C1} \uparrow \quad U_{C1} \downarrow$$

$$(U_{C1} = E_C - I_{C1} R_C; U_{C1} = E_C - I_{C1} R_C)$$

$U_{C1} \rightarrow$ base of T_2 transistor (starts blocking, so less current through T_2)

$$I_{C2} \downarrow \quad U_{C2} \uparrow$$

$$(U_{C2} = E_C - I_{C2} R_C)$$

$U_{C2} \rightarrow$ base of T_1 transistor (starts opening, so more current through T_1)

$$I_{C1} \uparrow \quad U_{C1} \downarrow \quad \text{Avalanche process}$$

Assume an initial state:

$$Q=1 \text{ and } Q\|=0 \Rightarrow$$

T_1 cut-off and T_2 saturated

Final state:

$$Q=0 \text{ and } Q\|=1 \Rightarrow$$

T_2 cut-off and T_1 saturated

Circuit operation:

- If current I_{C1} experiences a small rise, this leads to the decrease of the voltage drop U_{C1} , which is transmitted through the voltage divider R-r into the base of T_2 transistor. The decrease in the voltage drop U_{B2} will be amplified and inverted by the transistor, so the voltage drop U_{C2} will increase, and this increase will be transmitted in the base of the T_1 transistor, through the R-r voltage divider. Because of that, I_{C1} will increase even more.
- As a consequence, a switching process takes place, which develops like an **avalanche**: the I_{C1} current increases and the I_{C2} decreases, until T_1 becomes saturated, and T_2 off

State state is stable, because the positive reaction loop is interrupted, due to the blocking state of transistor T_2

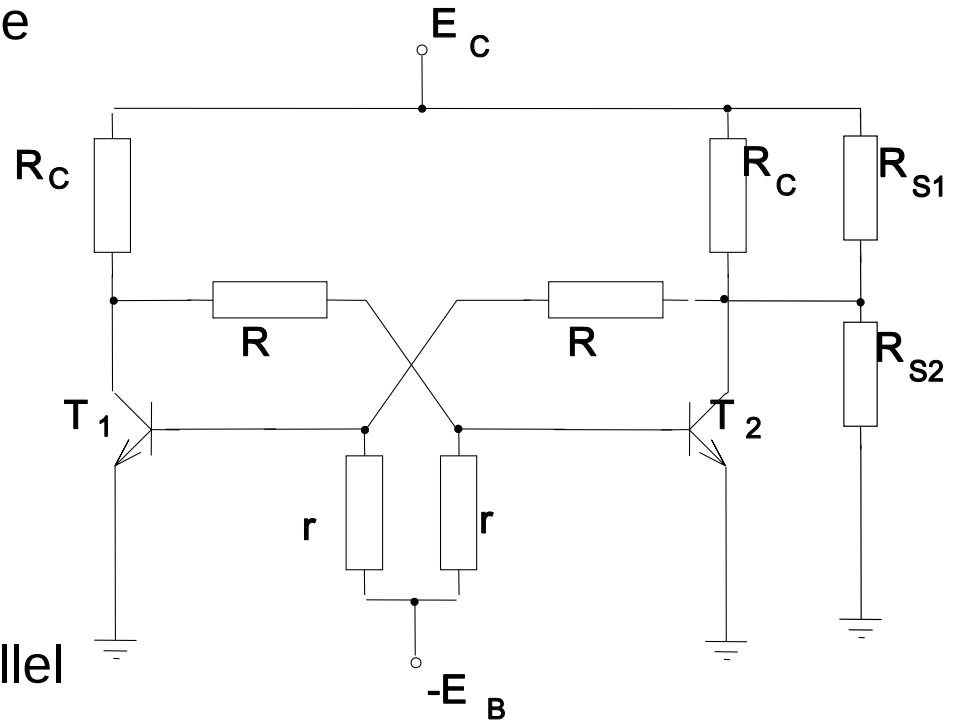
Two stable states:

1. T_1 – conducting (saturated), T_2 - blocked
2. T_1 - blocked, T_2 - conducting (saturated).

The Influence Of A Load Resistance On The States Stability

Connection of a load resistance at the output of a bistable may have a negative influence on the stability of the states, because it modifies the equivalent resistance of the collector circuit, affecting the distribution of voltages

Load resistance may be connected at one of the outputs, in parallel with the resistor R_C (case R_{s1}), or in parallel with the transistor T_2 (case R_{s2})



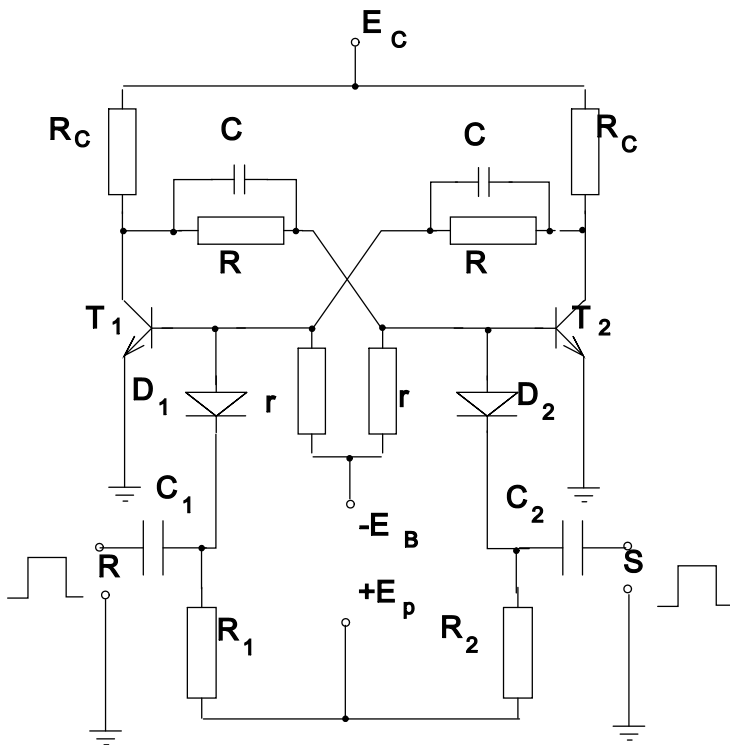
If load resistance R_{s1} is connected, equivalent collector resistance becomes:

$$R'_C = R_C || R_{s1} = \frac{R_C \cdot R_{s1}}{R_C + R_{s1}}$$

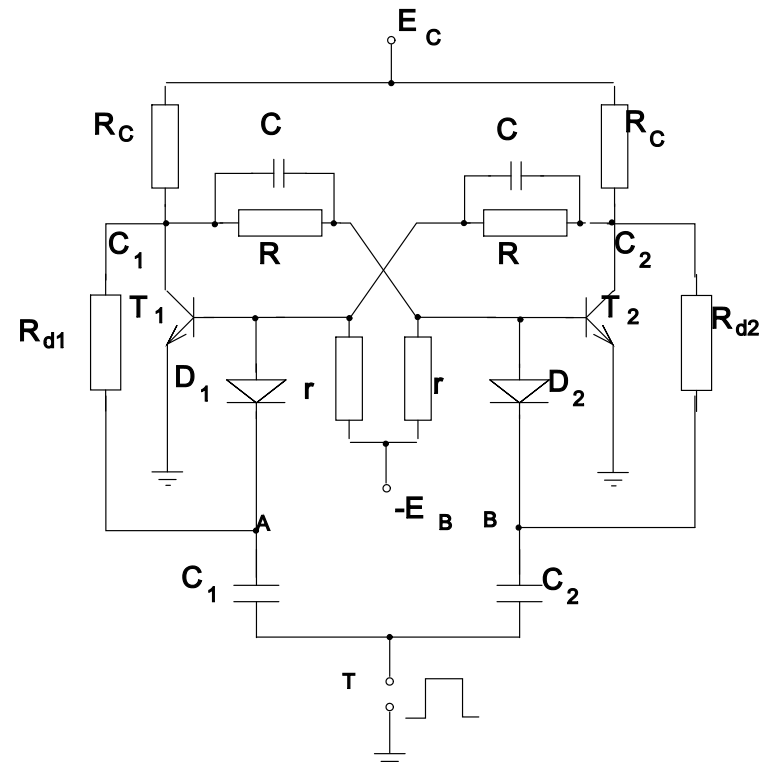
$R'_C < R_C$, so collector current is higher, driving base current must be higher

- In order to drive a flip-flop circuit with both pulses and voltage levels circuits usually have RC differentiating circuits.
- Through differentiation, positive and also negative peaks emerge, so, the possibility of unwanted double-switching exists.
- In order to avoid this situation, the differentiating circuits are followed by clipping diodes, which prevent peaks with unwanted polarity from passing through

Base-driven flip-flop circuit using separate paths

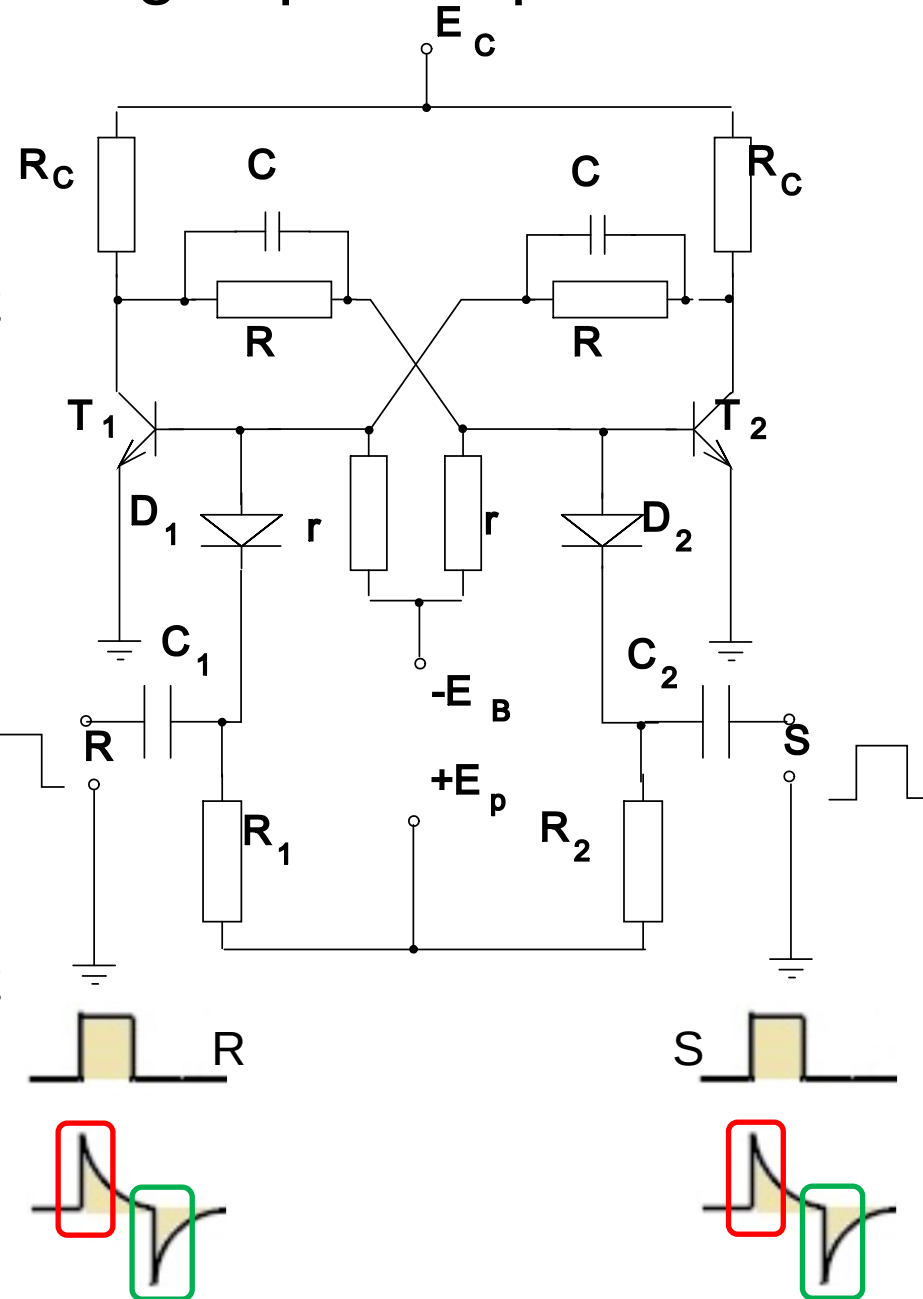


Base-driven flip-flop circuit using common path



Base-driven flip-flop circuit using separate paths

- Be the circuit in state T_1 - saturated and T_2 - off $\Rightarrow Q=0$ and $Q'=1$
- If a driving impulse is applied at the input S , having the amplitude E_c , it will be differentiated by the R_2C_2 differentiating circuit.
- Diode D_2 will cut the positive voltage peaks and permit only the negative peaks to pass
- These negative voltage peaks will block T_2 even more
- The switching does not take place in this case.
- If the driving impulse is applied at the input R , the negative voltage peaks, which reach the base of the T_1 transistor, will block it, causing the switching of the flip-flop circuit



- In order, for both **diodes** to be **blocked in stationary functioning regime**, it is recommended that they are both **reverse biased with a positive voltage E_p**
- Value of this voltage is chosen a little higher than the base-emitter voltage drop of the saturated transistor.
- There are cases in which a separate voltage source for biasing the diodes is not used, but rather a positive voltage obtained from a voltage divider, which is connected to the voltage source E_C . This way, regardless of the polarity of the driving impulses only the negative peaks will reach the base circuit. They will cause the switching of the circuit, if the corresponding transistor is conducting, having no effect otherwise.

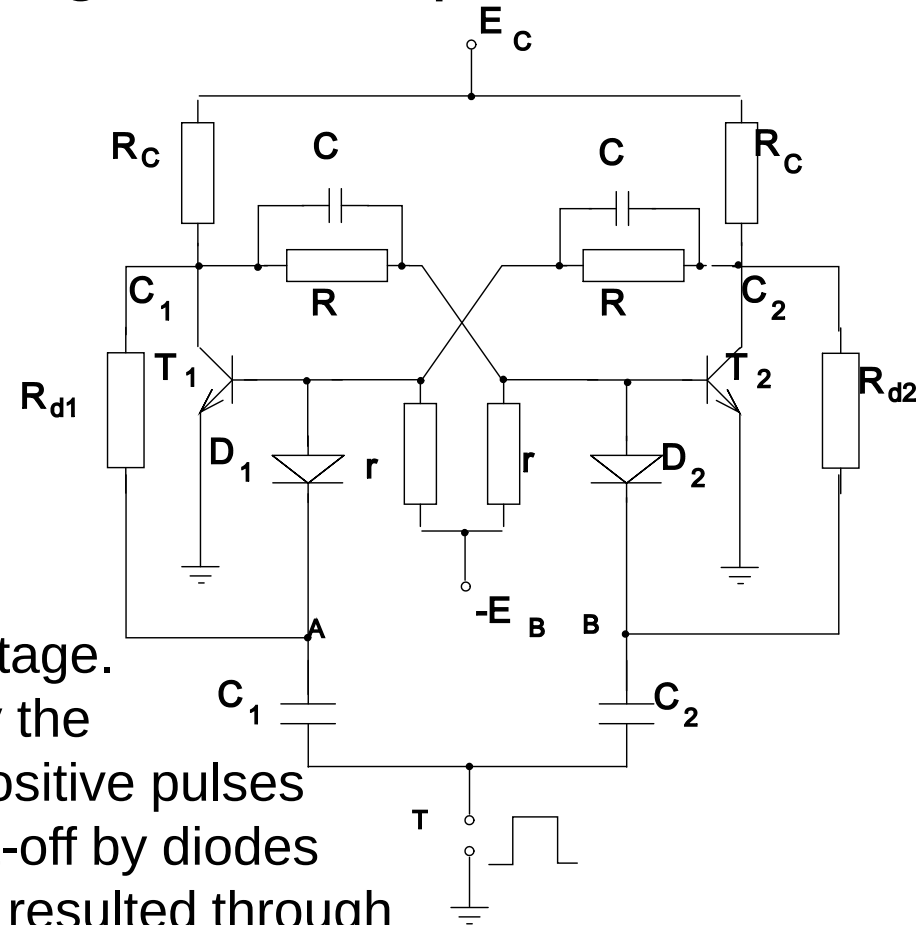
Collector-driven flip-flop circuit using separate paths

- Driving circuit is identical, the driving signal is applied on the collectors of transistors. Potential E_C is used as a positive bias voltage for diodes.

Base-driven circuits have some advantages, because allow for higher switching frequencies, and a higher sensitivity

Base-driven flip-flop circuit using common path

- Be circuit in state:
T1- saturated and T2- blocked.
- Voltages in different points of the schematics have the following typical values (for silicon transistors):
 $U_{C1} = +0,1V$; $U_{B1} = +0,7V$; $U_{B2} = -0,1V$
- D_1 diode is therefore conducting, and D_2 is blocked with a high reverse bias voltage.
- The driving pulse T is differentiated by the Groups R_{d1} , C_1 and R_{d2} , C_2 . The sharp positive pulses resulted through differentiation will be cut-off by diodes D_1 and D_2 . From the two negative pulses resulted through differentiation, only the one applied on the cathode of the conducting diode D_1 will pass through, while the other one will not be able to pass through the blocked diode D_2 .
- Driving signal will cause the switching of the circuit, because a sharp negative pulse is applied on the base of the conducting transistor



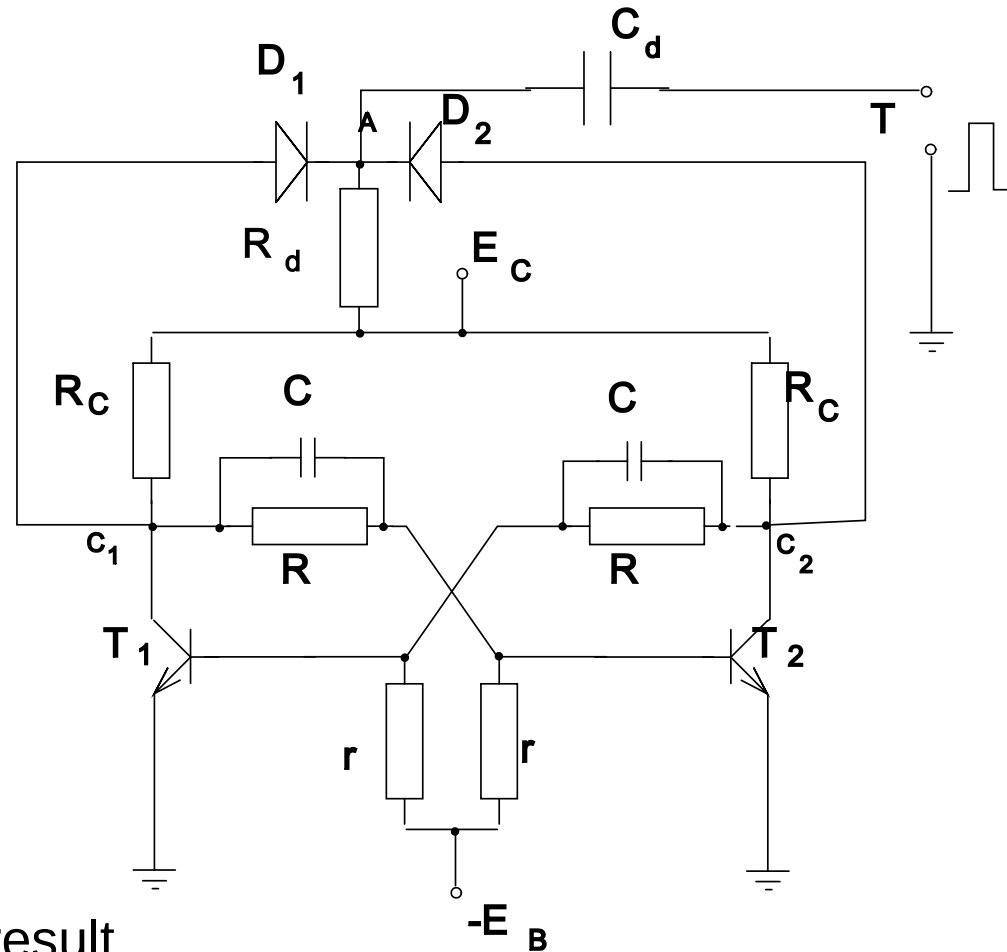
Collector-driven flip-flop circuit using common path

- Be the circuit is in state:
 T_1 - saturated and T_2 - off
- Voltages in different points:

$$U_{C1} = +0,1V; U_{C2} \gg +E_C,$$

$$U_A \gg E_C$$

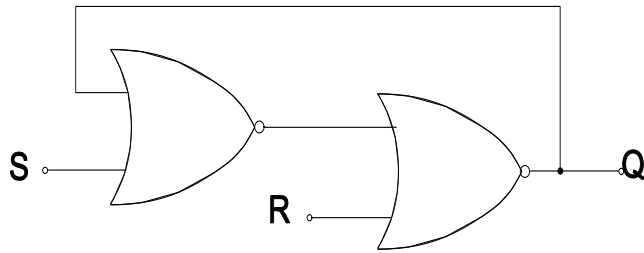
D_1 diode is blocked with a reverse bias voltage approx. equal to E_C , while D_2 diode is blocked with a small reverse bias voltage, equal to the voltage drop on the R_C .
 Driving pulse applied at the input T is differentiated by $R_d C_d$.



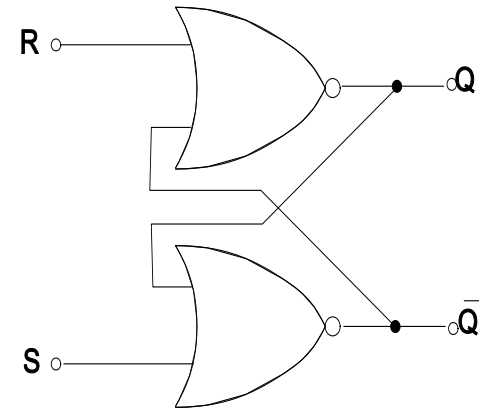
The negative voltage peaks, which result through differentiation (due to polarity of the diodes) can reach only the collector of the blocked transistor (the negative voltage leap is transmitted, in this case, through the D_2 diode in the collector of the cut-off T_2 transistor), making it to switch on.

Asynchronous S-R type flip-flop circuit

Design with gates (not discrete components)



RS type flip-flop with NOR gates

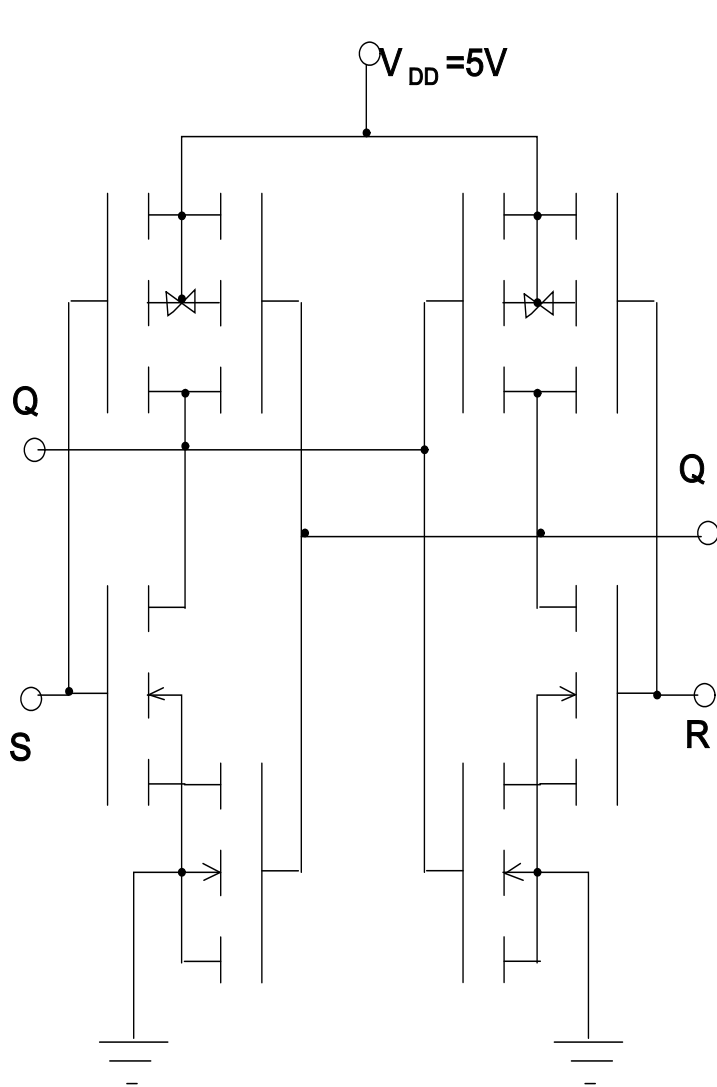


Symmetrical RS type flip-flop with NOR gates

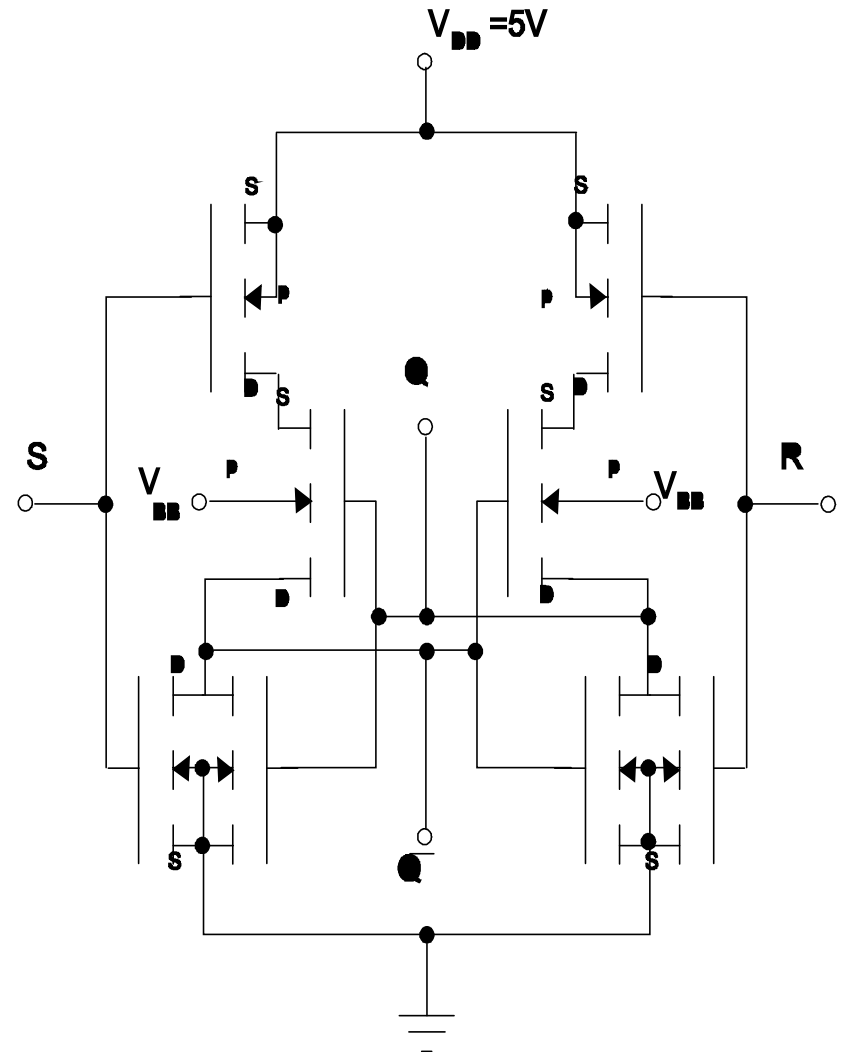
- Simple flip-flop circuit may be built, from a logical point of view, by introducing a reaction loop in a logic gate network, made of NOR or NAND gates.
- Inputs of the flip-flop circuit are called S (**set**) and R (**reset**)

S	R	Q^+
1	0	1
0	1	0
0	0	Q
1	1	Not allowed inputs

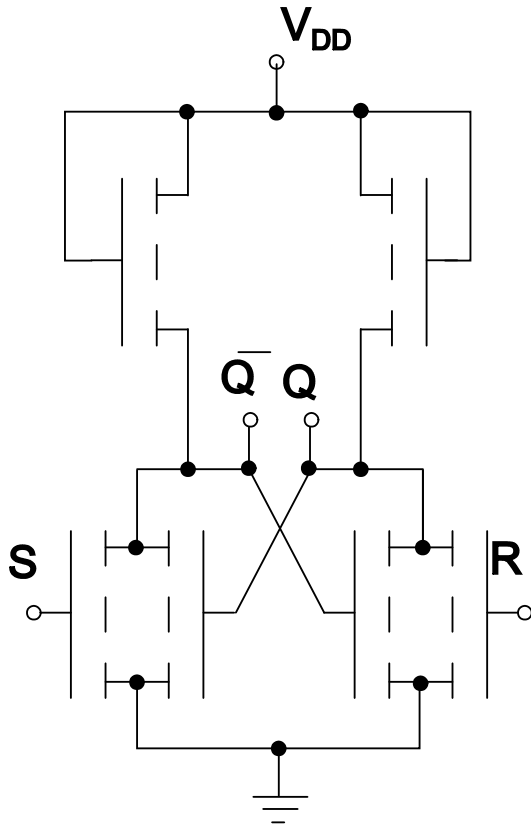
RS flip flop truth table



RS type CMOS integrated flip-flop circuit with NAND gates

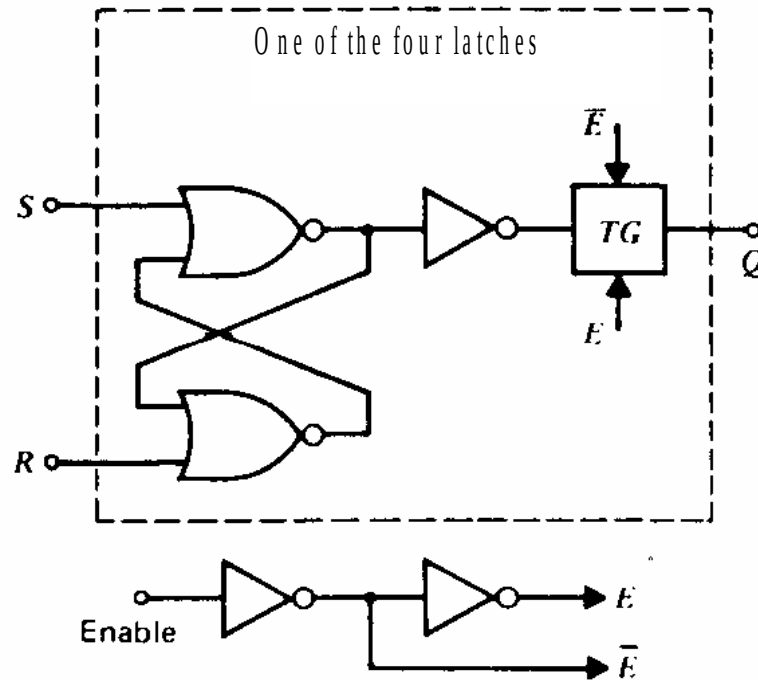


RS type CMOS integrated flip-flop circuit with NOR gates



Flip-flop with MOS components

An integrated circuit that contains RS type flip-flops with CMOS components: 4043 integrated circuit, which has four RS type flip-flops (*latches*)



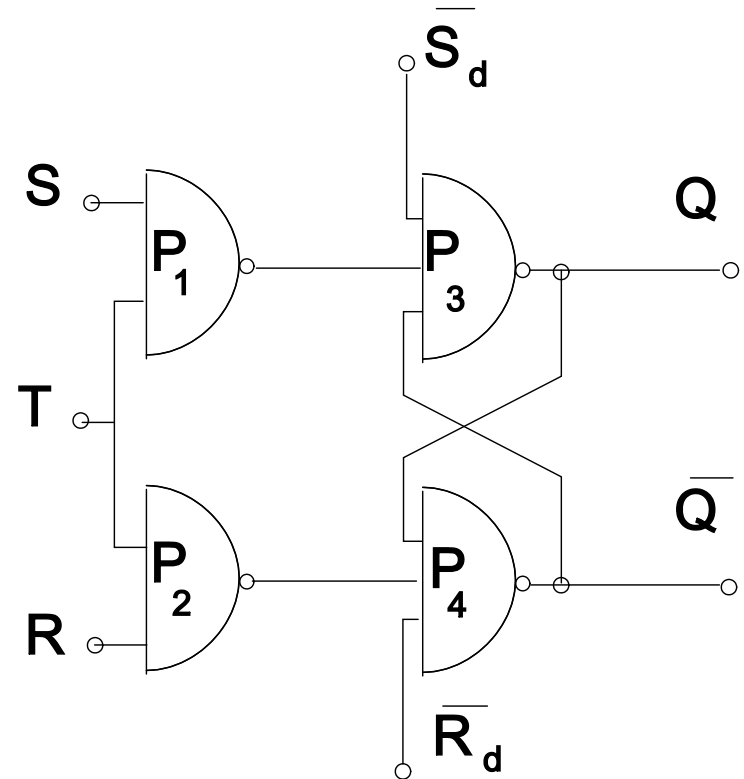
S	R	E	Q
X	X	0	OC
0	0	1	NC
0	1	1	0
1	0	1	1

OC = Open circuit
 NC = No change
 X = Indifferent value

The internal logical schematics for the 4043 circuit

Synchronous SR type flip-flops

- Synchronous RS type flip-flops have two data inputs: R and S and a clock input: T.
- The information from the data inputs R and S can be transmitted to the flip-flop itself only when the clock pulse has the logic value '1'.



- The synchronous RS type circuit has a clock signal (the inputs R and S remain asynchronous), which controls the evolution of the circuit.
- It also has two other inputs, **Clear** and **Preset**, which act directly on the outputs Q and Q- overriding the inputs R and S

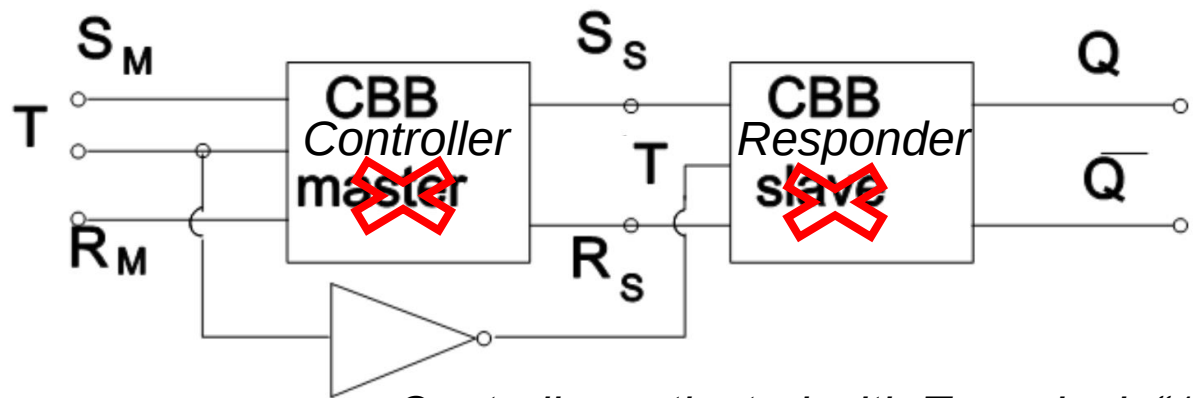
Problem: possible non-determination of the states for each flip-flop

- *as long as T is in logic one, R and S can be activated several times
? what value should we pass to the next flip-flop ?*

Solution: Only one value (the last should be good)

RS master-slave flip-flop

RS controller-responder flip-flop



Controller activated with T on clock “1” and Responder with second T on clock “0”

- **RS master-slave** flip-flop: designed to overcome the disadvantages of cascading several RS type flip-flops (the possible non-determination of the states for each flip-flop)
- First flip-flop, called master flip-flop is driven by the data inputs R and S, while the second flip-flop, called slave flip-flop, is being driven by the outputs of the master flip-flop. Short description:
 - on the rising edge of clock pulse, the master flip-flop is disconnected from the slave flip-flop, as they can't communicate; On this edge, the S and R inputs act on the master flip-flop, determining the corresponding switching
 - when the clock signal goes from '1' into '0', on the falling edge, the slave RS inputs are disconnected from the master; the outputs of the master drive the state of the slave.
- This way, **only one single flip-flop is active at a given moment, the outputs of the master-slave flip-flop being completely isolated from its inputs.**
- Using this circuit, logic uncertainty for a sequence of flip-flops is avoided.

Positive feedback circuits: **MULTIVIBRATOR CIRCUITS**

Bistable

Monostables

Astables

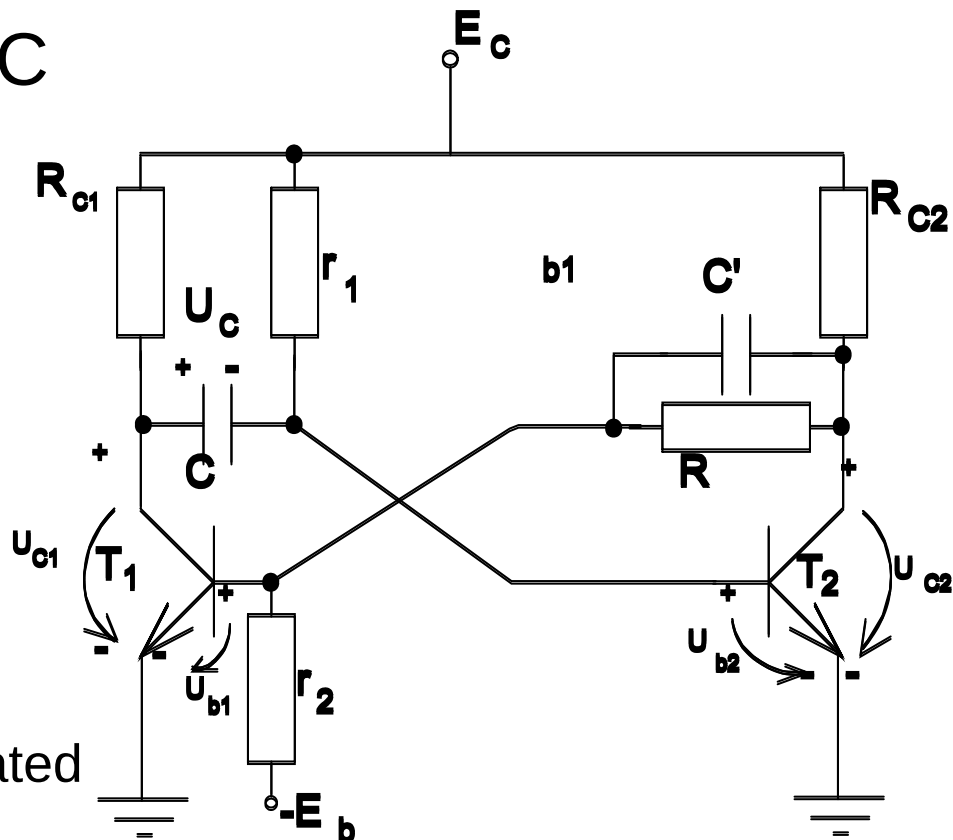
Monostables circuits

- Monostable switching circuits (MSC) present a **stable** and an **unstable** state; are used especially for obtaining some fixed time intervals, precisely marked by voltage swings
- Most important condition for these circuits is the **duration** for the **unstable state**

Collector-base Coupled MSC

- Positive reaction is assured by coupling circuits, but at MSC the resistive coupling between the collector of T_1 and the base of T_2 is replaced by a capacitive coupling.
- The different nature of states is due to the asymmetry of the coupling

Stable state: T_1 - cut off and T_2 - saturated



Stable state: the transistor T_1 is cut off, because its base is connected through r_2 to $-E_B$, and the transistor T_2 is saturated, because its base is connected through r_1 to E_C .

The condition that T_1 be cut off can be written analogically with CBB: $r_2 \leq \frac{E_B}{I_{C0\max}}$

Condition for T_2 to be saturated is: $\beta_{\min} I_{B2} \geq I_{C2}$ or

$$\beta_{\min} \frac{E_C}{r_1} \geq \frac{E_C}{R_{C2}} \quad \text{resulting value:} \quad r_1 \leq \beta_{\min} R_{C2}$$

Voltage fall on the capacitor C , U_C will be for this stable state $\approx E_C$

Unstable state:

By applying a negative pulse on the base of the transistor T_2 , it is initialized a switching process, which can be described like this:

- the voltage U_{B2} decreases
- the collector current I_{C2} decreases
- the potential in the collector of T_2 (U_{C2}) increases, same for the potential U_{B1}
- the collector current I_{C1} increases
- the potential in the collector of T_1 , U_{C1} , decreases, same as U_{B2}

In this way the circuit passes to the opposite state:

T_1 is saturated and T_2 is cut off

Due to the fast fall down of the voltage U_{C1} from E_C to approximately 0V, it will make the capacitor C to transmit a negative voltage swing, of amplitude E_C , to the base of T_2

After this transition: $U_{C1}(t_1) \approx 0$, $U_{C2}(t_1) \approx \frac{R}{R + R_{C2}} E_C$

$$U_{B2}(t_1) \approx U_{B1}(0) - E_C \approx -E_C$$

During the unstable state, capacitor C starts discharging from initial value: E_C to final value: $U_C = -E_C - I_{C0}r_1$ and voltage U_{B2} will grow exponentially to $-U_C$. This positive voltage will open T_2 and due to positive reaction, circuit will be back in the stable state

Duration of the unstable state, given by analysis of the blocking condition of transistor T_2 and the saturation condition of T_1 .

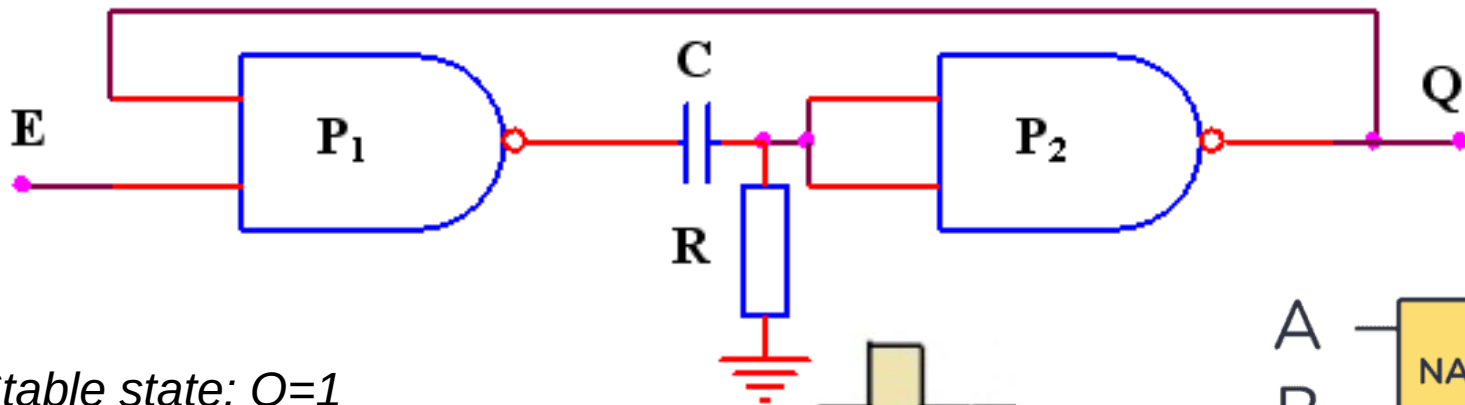
Results the dimension of the resistance R:

$$R \leq \frac{\beta_{\min} E_C}{\frac{E_C}{R_{C1}} + \frac{2 E_C}{R_B} + \frac{E_B \beta_{\min}}{r_2}} - R_{C2}$$

Simplified time duration for unstable state: $T_{si} \approx \tau \cdot \ln 2 = 0,69 \cdot r_1 \cdot C$

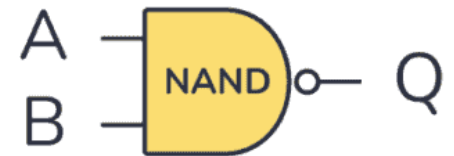
Monostable circuit using TTL gates

- Circuit built up with two TTL NAND gates and a RC circuit
- The switching will be triggered by low active enable signal E



Stable state: Q=1

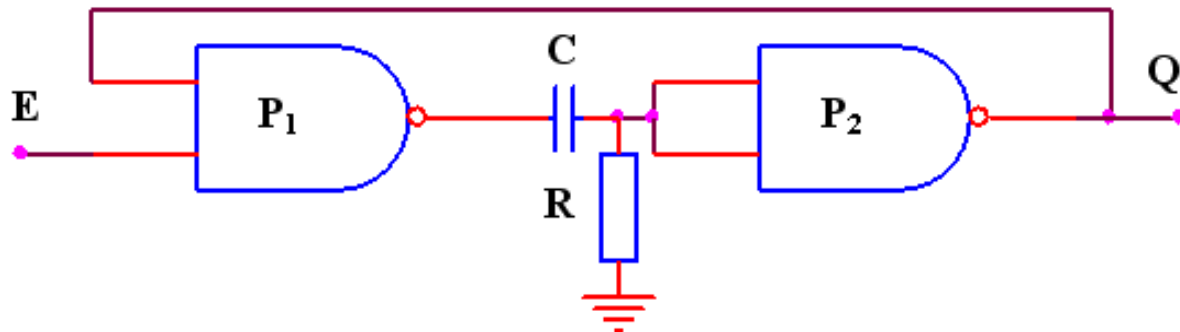
- apply on E = 0
- P1 = 1 => pulse on HPF
 - Positive spike -> P2 = 0 => Q=0 (P1 will not switch)
 - Negative spike -> P2 = 1 => Q=1 (P1 = 0)



A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

- When applying E, at the output of the first gate P1 will exist a positive jump which will be transmitted through the capacitor C to the input of gate P2, which initially was biased at the stable state through the resistance R, connected to ground.
- When the positive jump appears on the input of the gate P2 this will switch, the negative voltage jump from its output will be transferred to the input of the gate P1, which will remain in '1' logical level;
- In time, through gate's output transistor, the capacitor C will discharge from V_{CC} to the ground.
- When on the input of the gate P2 the potential will reach the threshold, the gate will switch back to '1' level, and together with the end of the negative pulse, will make the gate P1 to pass into '0' logical level, the stable state.
- The approximate duration of the unstable state is:

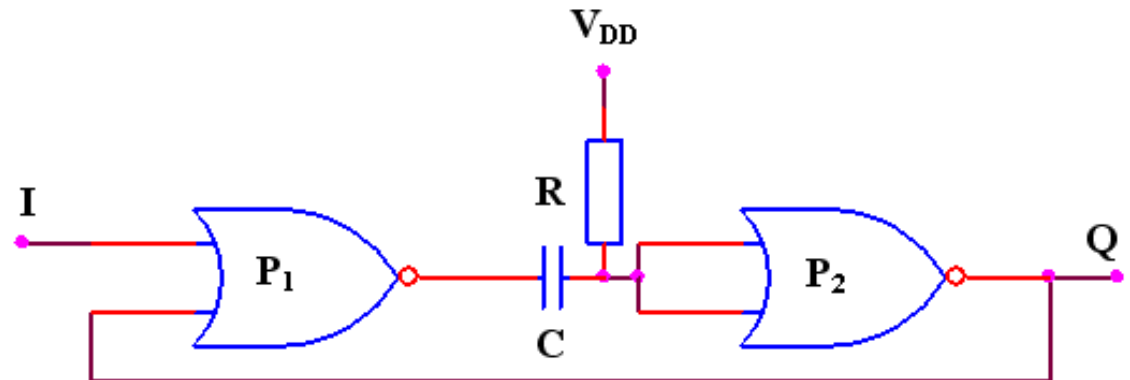
$$T_{si} \approx RC \ln \frac{V_{OH} + V_{IL} - V_{OL}}{V_T}$$



Monostable circuit using CMOS gates

- Circuit built up with two CMOS NOR gates and a RC circuit
- The switching will be triggered by high active enable signal I
- Stable state will make output Q at '0' due to R connected to V_{DD}
- When at the input I is applied a positive triggering pulse, the gate P1 will present at the output a negative voltage jump, which will reach the input of the gate P2 through the capacitor C, producing the switch of this to '1' logical level.
- Then the capacitor C will charge up from V_{DD} and when the potential on its terminals will reach the switching threshold value, the gate P2 will switch back to '0', the stable state of the circuit.
- The duration of the unstable state can be determined with the formula:

$$T_{si} \approx RC \ln \frac{V_{DD}}{V_{DD} - V_T}$$



Positive feedback circuits: **MULTIVIBRATOR CIRCUITS**

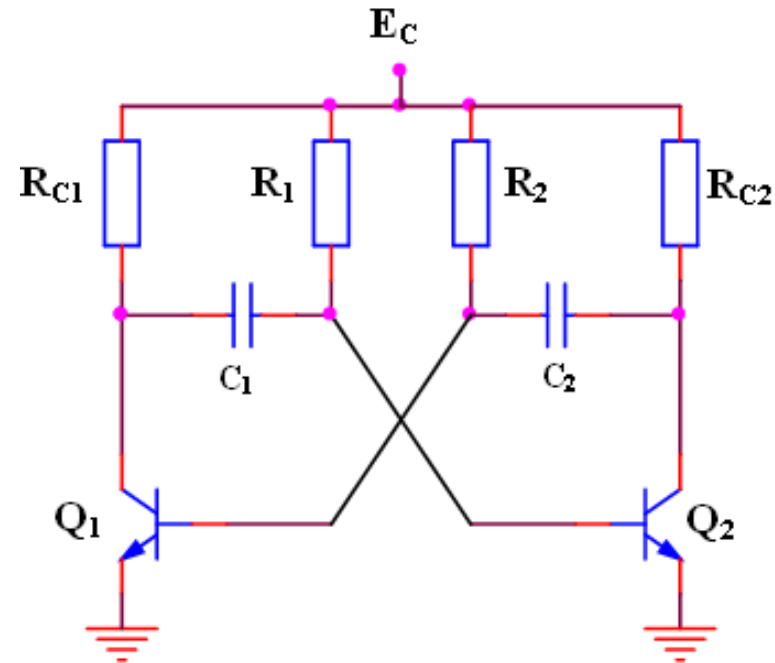
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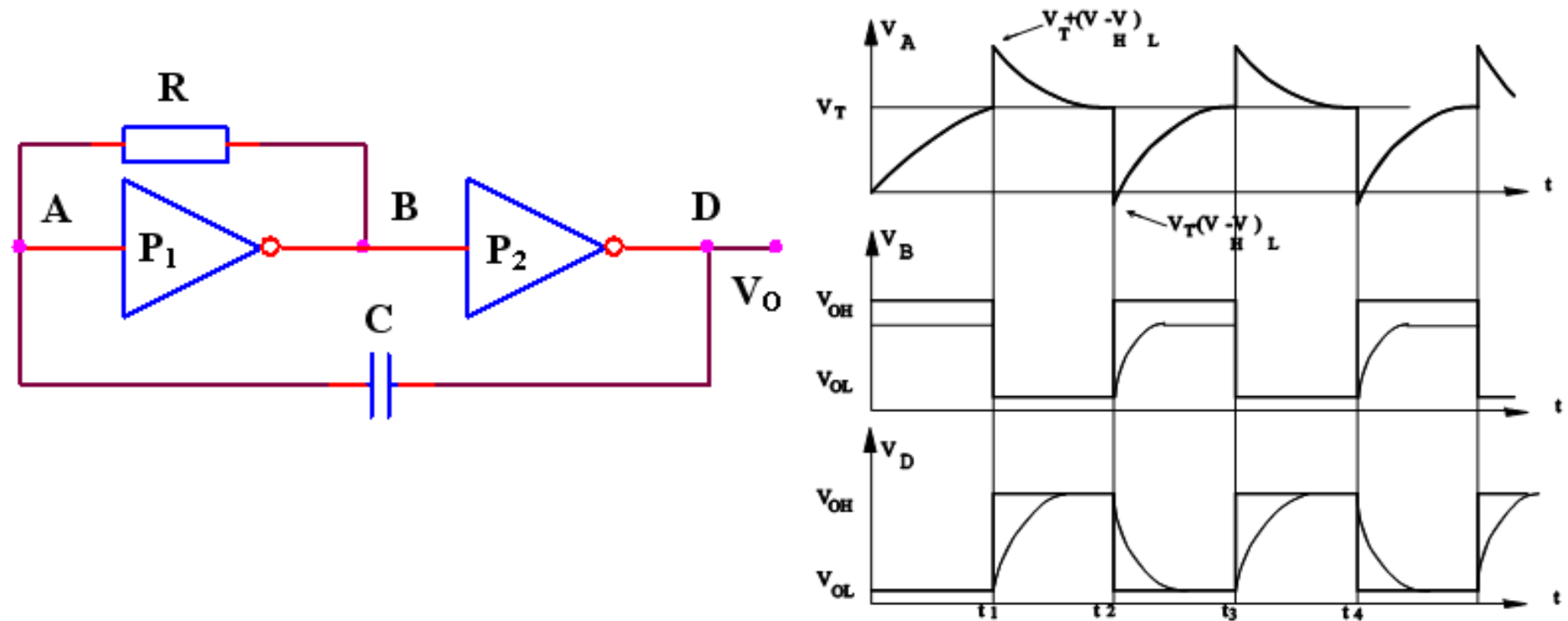
ASC with discrete components

- Existence of **two unstable states**.
- The switching is executed in a very short time and appears with a sudden variation of the electrical quantities.
- **Switching** is triggered **without applying any signals from outside**;
- Duration of states defined by the circuit's parameters.
- The **ASC is an oscillator**, which produces at the output a square signal.
- The duration and the period of the impulses are determined by some elements of the circuit.
- Since R_1 and R_2 resistances are tied to the source $+E_C$, assures that transistors in stationary regime will not remain blocked for an undefined period of time



ASC with collector-base coupling illustrated

ASC implemented with TTL gates



Main idea: the value stored on C will drive A to 1 or 0 (over/under V_T of the gates)

Initial state: $A=0 \Rightarrow B=1$ and $D=0$

Due to capacitor C charging up (through resistance R from high voltage in B) A tends to increase to V_H .

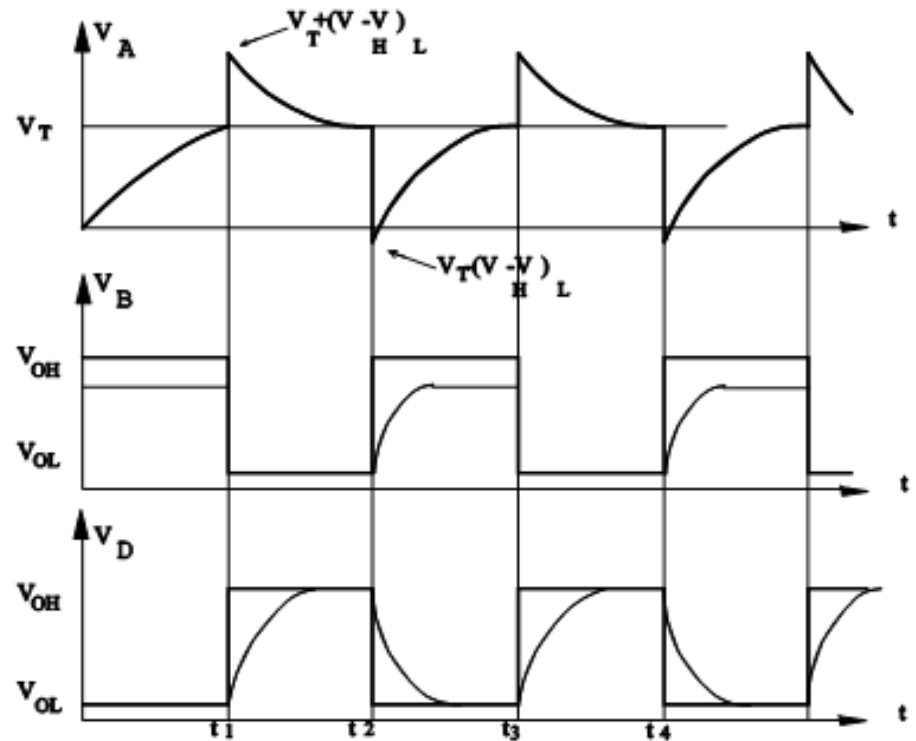
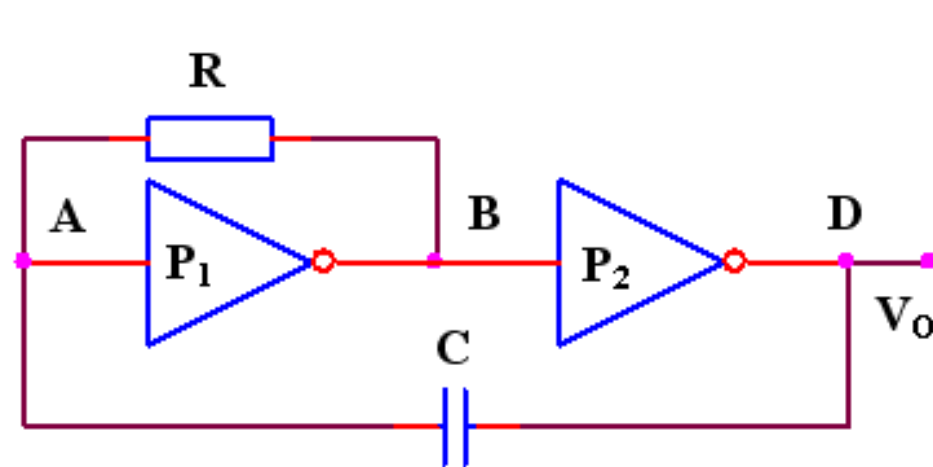
At the moment $t=t_1$ when $V_A=V_T \Rightarrow P_1$ switches $\Rightarrow B = 0$ and P_2 switches $\Rightarrow D = 1$

Discharge capacitor C (through R toward low potential of B) $\Rightarrow$ A will start decreasing exponentially to the low

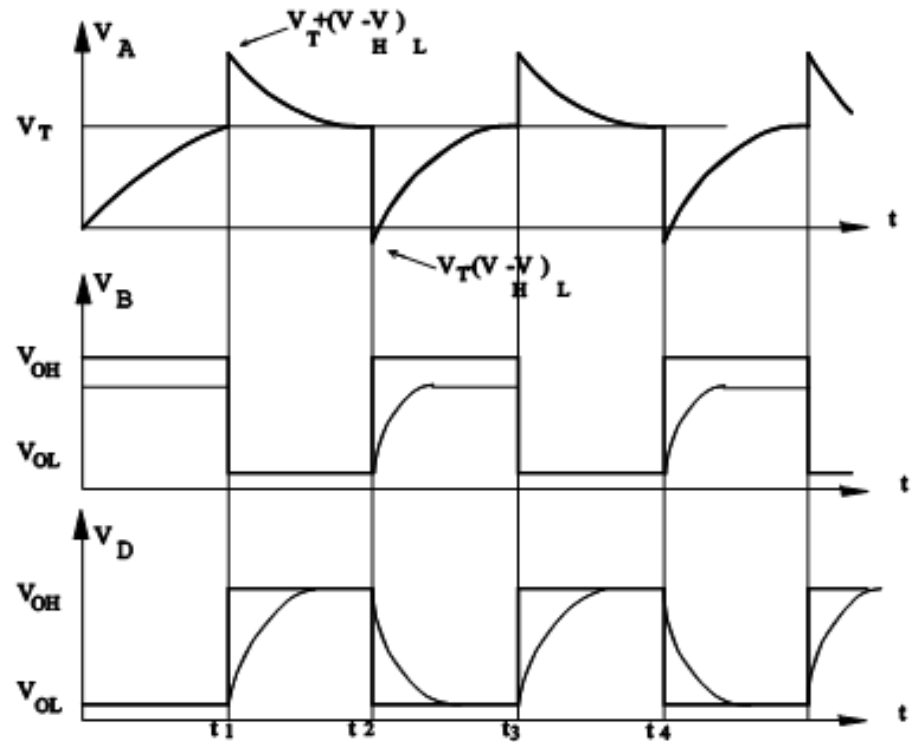
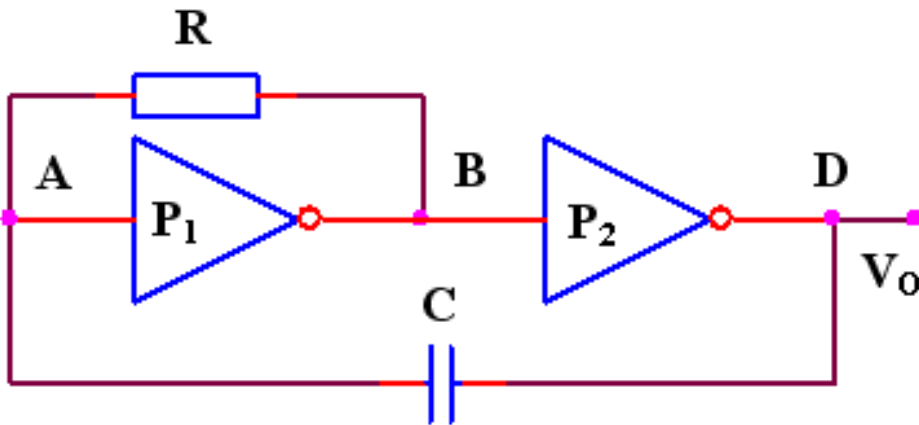
At the moment $t=t_2$, $V_A=V_T \Rightarrow P_1$ switches $\Rightarrow B = 1$ and P_2 switches $\Rightarrow D = 0$

$\Rightarrow$ C will charge up through the resistance R, and the potential from A will increase again

ASC implemented with TTL gates

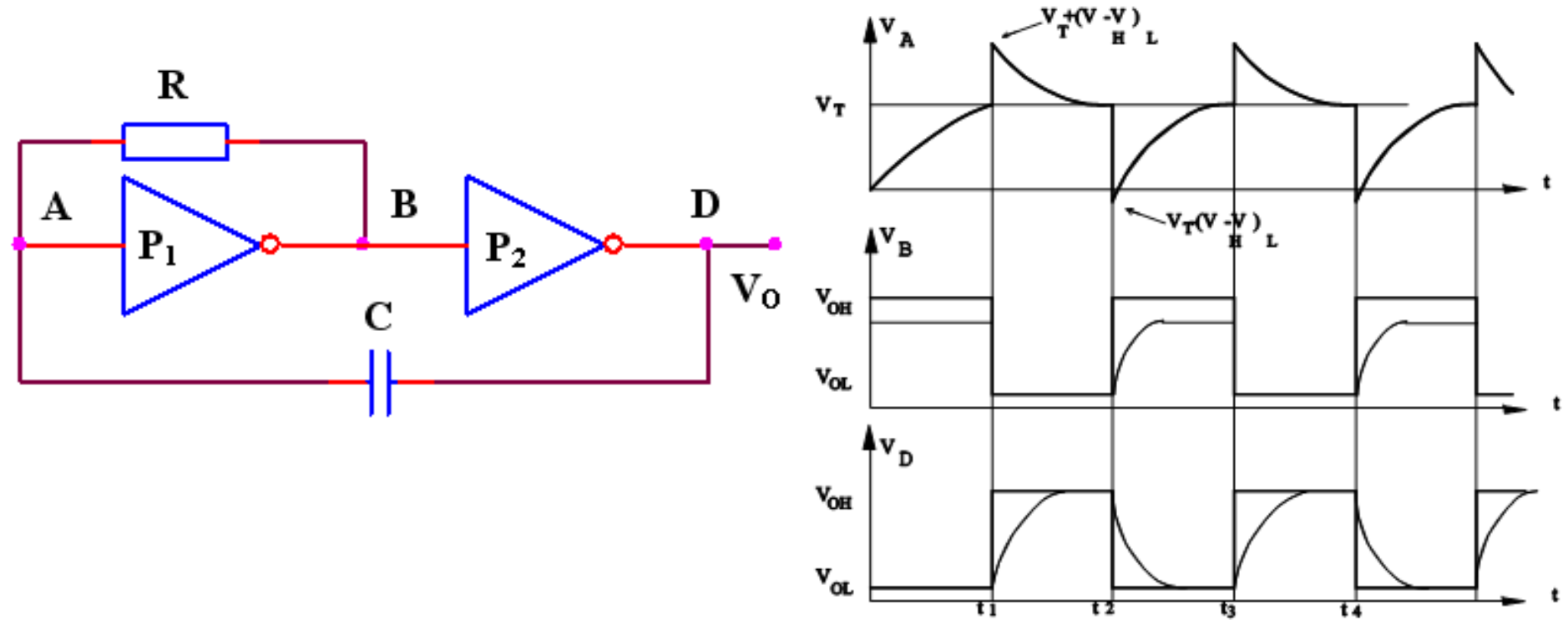


- Operation: if at moment $t < t_1$ at the input of the gate P_1 (A point) there is logical '0' signal, at its output (point B) will be logical '1', and at the output of the gate P_2 will have logical '0'. So, potential of the point A tends to increase to V_H , due to capacitor C charging up through resistance R from high voltage in B.
- At the moment $t = t_1$ when $V_A = V_T$, (threshold potential of P_2), the voltage in D goes high; this jump will be transmitted through capacitor C to point A.



- The voltage from point A will start decreasing exponentially to the low value from the output of the gate P_1 , during the discharge process of the capacitor C through R toward low potential of B.
- At the moment $t=t_2$, $V_A=V_T$ which determines again the two gates to switch.
- The voltage jump from the point D will be transmitted through the capacitor C to A. Then the capacitor will charge up through the resistance R , and the potential from A will increase...
- The phenomenon will continue until the circuit is powered

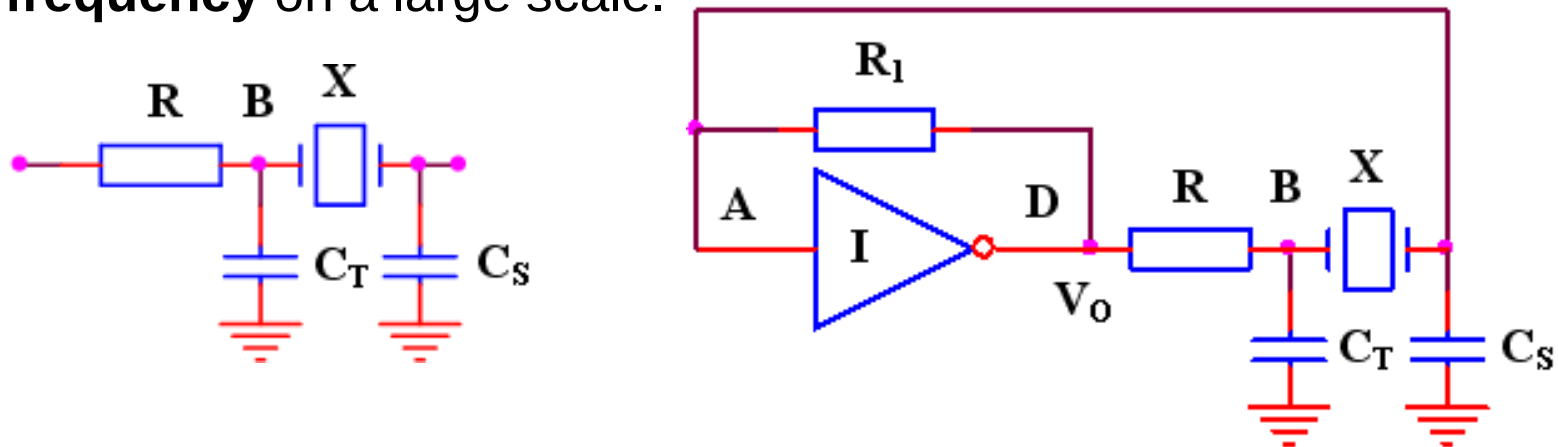
ASC implemented with TTL gates



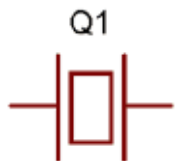
- Problem of ASC with TTL
 - have low precision - it depends on the power supply, changes in temperatures, R C elements that have a tolerance in value of +/-%, capacitors can lose their value/properties over time
- Solution:
 - external timing

ASC implemented with quartz crystals and logic gates

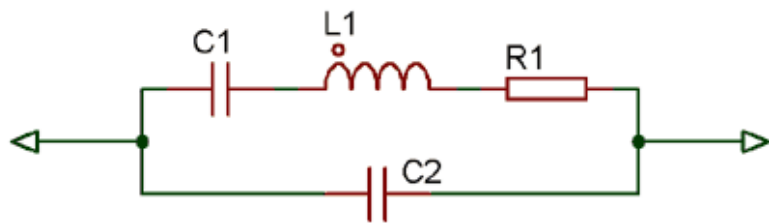
- Quartz is cut in certain crystallographic directions, sanded and metal electrodes are deposited on two parallel faces. From an electrical point of view, the crystal provides an impedance with resonant circuit properties.
- Oscillators with quartz crystals realized with CMOS circuits assures the advantage of the **low power consumption** and the **stability of frequency** on a large scale.



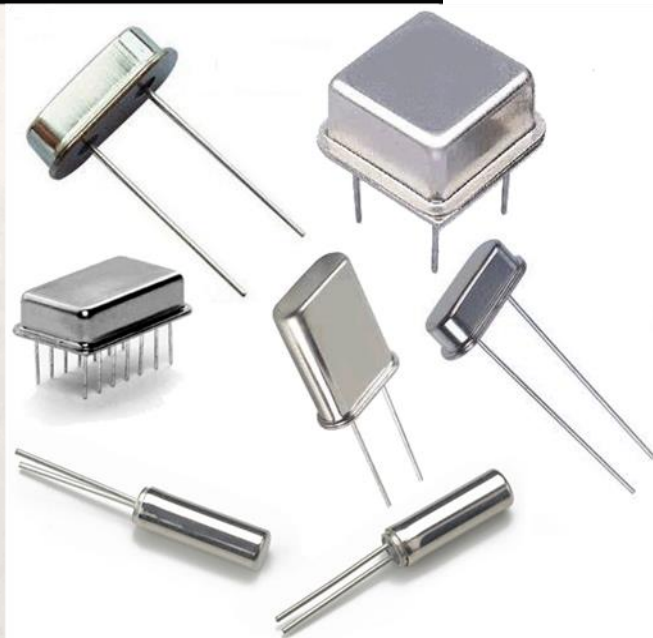
- Fundamental oscillator contains an amplifier and a network with reaction. The circuit presented in figure, also called π network with quartz, is indicated to be used together with an amplifier, which assures a 180° phase shift.



Symbol of Quartz Crystal



Electrical Equivalent of Quartz Crystal



https://www.youtube.com/watch?v=sc4EFAqZgPU&ab_channel=EpsonAmerica

https://www.youtube.com/watch?v=lzHqhNoyx2o&t=2s&ab_channel=NikkeiAsia

Steve Mould - piezoelectricity

Proposed Problems

Design a monostable circuit with CMOS gates, presenting an unstable state of 0,1ms.

Determine the unstable state duration for a monostable circuit based on TTL gates, if: $R=220\Omega$ and $C=47\text{nF}$.